

FILTER DESIGN TEST CASE

Low-Pass Filter - Band 12 (700 MHz) for WiFi 6E Coexistence Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting Band 12 (700 MHz) for WiFi 6E Coexistence Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 1754.3 MHz and bandwidth of 81.6 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: Band 12 (700 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: 128° YX LiNbO3
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Low-Pass Filter design targeting Band 12 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1754.3 MHz
Bandwidth (3 dB):	81.6 MHz
Insertion Loss Target:	≤ 1.2 dB
Return Loss Target:	≥ 12.0 dB
Out-of-Band Rejection:	≥ 27.0 dB
Isolation (Duplexer):	≥ 24.0 dB
Q Factor:	8862
Temperature Coefficient:	-23.6 ppm/°C
Die Size:	2.8 x 2.2 mm
Wafer Size:	8-inch
Number of Resonators:	7
Electrode Material:	W

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Low-Pass Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 1754.3 MHz, BW 81.6 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 5263 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 122 MHz offset >= 27.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 5.2 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2025-01-02
Devices Tested: 38
Insertion Loss (meas): 0.92 dB
Return Loss (meas): 13.8 dB
Rejection (meas): 30.0 dB
Isolation (meas): 28.6 dB
Q Factor (meas): 8862
Group Delay Variation: 3.8 ns
Power Handling: +27 dBm
Die Yield: 94.7%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated acceptable performance for Band 12 (700 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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